

ISONOMIA

Executive Summary

*Open-source software for storing, citing, and checking the reasoning behind a conclusion.***

Isonomia turns an argument into structured, verifiable data instead of prose. Today the reasoning behind a decision lives buried inside documents — a web page mixes the claim, the evidence, the rhetoric, and a great deal of unstated assumption into one blob, and if you want to cite it, you cite the whole page. Isonomia breaks that apart. A claim or an argument becomes its own object with a permanent address, carrying what supports it, the sources behind it (fetched, timestamped, and verifiable, so the record holds even if the original link later rots), the strongest objection on file against it, and whether it has survived challenge. The result is something a person — or, increasingly, an AI system — can cite precisely, trace to its origin, and check, rather than re-reading and re-judging a document every time.

Why it should exist. Reasoning is the one thing software has never given a durable home. We store documents, messages, transactions, and code, but the inferential structure that connects evidence to a claim to a conclusion is thrown away the moment a decision is made — left to be reconstructed, badly, from prose every time someone asks *why*. That gap is becoming expensive at exactly the moment it is becoming unavoidable. AI systems are moving from answering questions to acting across tools, and they need a place to read and write reasoning *state* — what has been asserted, what supports it, what attacks it, what survived, and what a conclusion does not yet license — that lives outside any single model, where it can be inspected, versioned, and contested. Better models do not remove this need; they raise it, because the cheaper it becomes to generate a confident claim, the more valuable it becomes to have an external, accountable record of which claims have actually earned standing. Isonomia exists to be that record: the missing substrate that makes reasoning a first-class, persistent, checkable object rather than a disposable byproduct of writing.

The problem it addresses. As language models produce a growing share of what we read, the bottleneck is no longer fluent text; there is effectively infinite fluent text now. The bottleneck is trust: can you tell where a claim came from, whether its source actually supports it, and what it leaves out? Models are now good at writing about particular topic-based discourses and bad at keeping their structure straight. They flatten which point is doing the real work, miss objections aimed at the logic rather than the conclusion, and cite a page when the meaningful unit is a single sentence. And asking one model to grade another's reasoning yields a verdict that shifts with the wording. The academic field that studies how to represent arguments formally is called computational argumentation; Isonomia applies it to exactly this gap — keeping the structure of reasoning outside the model, where the things that can be checked are checked rather than guessed.

How it works. The structure is captured inside the work where reasoning is already happening — research, document review, policy analysis — rather than bolted on afterward as a separate annotation chore, which is precisely where earlier attempts at this broke. AI can propose the first-pass structure; people confirm or correct it where the commitment matters; and the system records who authored what, so machine-generated material stays visibly provisional until a human ratifies it. Everything is exposed both as ordinary web pages and over the same kind of connection AI assistants already use to reach outside tools, so a search engine, a researcher, and an automated agent all read the same object. Every citation carries its strongest known counter-argument by default, which makes one-sided citation harder to do by accident.

Where it leads. The single object is the seed of larger infrastructure, and each next step is an extension of what already runs rather than a pivot. As the objects accumulate, they form a *web of machine-citable arguments* — a public corpus where every claim carries its support, its opposition, its provenance, and its standing, addressable the way papers are addressable today. That corpus becomes a reasoning backend an agent can call: the service it queries to learn whether a claim has survived challenge, where a disagreement actually lives, and what it would have to retract to reject a conclusion — so that an AI writer can be handed not just sources but a contract for what it may not assert and what it must hedge. The same substrate yields a *minimal-disagreement debugger* that locates the single point where two positions first diverge instead of declaring them globally opposed; a transport layer that moves arguments between communities and institutions with their provenance intact; and an evidence algebra that folds confidence lawfully rather than as an opaque score. The reasoning record also becomes training and evaluation data no one else holds — arguments with their premises, schemes, objections, and resolution trajectories tracked over time. The compounding asset is not any one feature but the accumulating graph of reasoning itself.

Who it's for. The value concentrates wherever the reasoning behind a decision matters as much as the decision itself and has to survive scrutiny later: research groups, peer review, policy and regulatory analysis, and the teams evaluating AI systems. Alongside those, AI tools themselves are a natural user — a model consuming structured arguments as a citation source does not tire of structure the way a human volunteer does, and an agent that writes back into the graph extends a shared record rather than spending its work into prose that scrolls past. A broad public version is the long-term upside, not the starting point. The platform is open-source, self-hostable, and free; the communities that produce the reasoning own it and can carry it out in open formats. What is being built is not a better way to write about a debate but the place the debate is kept — the epistemic substrate underneath research, writing, search, and increasingly the AI systems doing all three.